University of Dhaka
Department of Software Engineering

Flood Detection
using Satellite Image Processing





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A Project for
Software Engineering
University of Dhaka

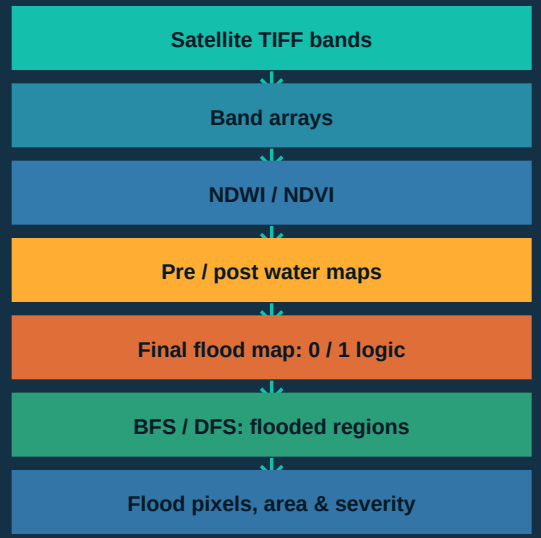
Detect • Analyze • Mitigate

Project Overview

Using multispectral satellite data to identify surface-water patterns and build toward flood mapping.

- Flooding can rapidly change the extent of surface water across a landscape.
- Satellite imagery provides repeat observations of the Earth's surface.
- This project uses Sentinel-2 optical bands and spectral indices as an initial water-mapping workflow.
- The current midterm milestone is to read, validate and process two dated image sets and generate separate water maps.

Processing pipeline



MIDTERM PRESENTATION

Work Completed at a Glance

The processing sequence completed up to the midterm stage.

01 Sentinel-2 data

Collected 8 single-band TIFF files.

02 Image reading

Read the four bands for both dates into program memory.

03 Input validation

Checked dimensions, file opening, grid data and CRS.

04 Preprocessing

Checked pixel values and NDWI denominator validity.

05 Band storage

Stored all 8 band arrays for both acquisition dates.

06 Index calculation

Calculated NDWI and NDVI for each date.

07 Water maps

Generated pre-flood and post-flood water maps.

2 acquisition dates • 4 bands per date • 8 input files

Sentinel-2 Dataset and Selected Bands

Two acquisition dates are currently used: 09 January 2019 and 14 January 2019.

- Each date contains four single-band GeoTIFF files.
- B02 – Blue | B03 – Green | B04 – Red | B08 – Near Infrared (NIR).
- The four selected bands support visible/NIR analysis and the NDWI/NDVI calculations implemented so far.
- Each raster contains 512 × 512 pixels (262,144 pixels per band).

Sentinel-2 satellite images



PRE-FLOOD | 09 JAN 2019

POST-FLOOD | 14 JAN 2019

Reading Sentinel-2 Images

The selected single-band images are read and their pixel values are loaded into memory.

IMAGE INPUTS

B02

BLUE • 512 × 512 pixels

B03

GREEN • 512 × 512 pixels

B04

RED • 512 × 512 pixels

B08

NIR • 512 × 512 pixels

READING PROCESS

- Open each input image file
- Read its single raster band
- Load pixel values into arrays
- Repeat for both acquisition dates

8 image files loaded

4 bands × 2 dates

Input Validation

The input images are checked before preprocessing and index calculation.

01 — Image dimensions

Confirmed the expected height and width (512 × 512).

02 — File opening

Checked whether every input file opens successfully.

03 — Georeferencing information

Checked raster origin and pixel-size information.

04 — Grid data

X origin · Y origin · Individual pixel height · Individual pixel width
X rotation · Y rotation

05 — Coordinate reference system

Checked the CRS information of the input images.

INPUT VALIDATION · ILLUSTRATION

Raster metadata & grid inspection



GEOREFERENCING

- X origin
- Y origin
- Pixel width
- Pixel height
- X / Y rotation

Raster grid

512 × 512 cells

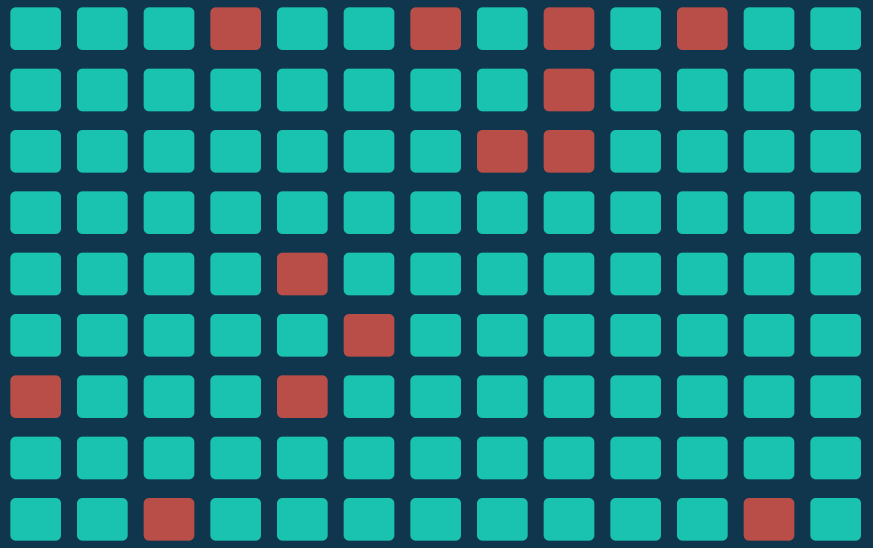
Illustrative validation view · dummy graphic

All input checks are completed before further processing.

Pixel Value & Denominator Checks

Preprocessing checks the input values before spectral-index calculation.

PIXEL-BY-PIXEL CHECK



Illustrative validity scan

PREPROCESSING CHECKS

- Checked whether every pixel value is valid across the input bands.
- Checked whether the NDWI denominator (Green + NIR) is equal to zero.

Band Values Stored in Arrays

A sample pixel demonstrates that values from all bands and both dates are accessible.

PROCESSING VISUAL

ILLUSTRATIVE · NOT ACTUAL OUTPUT

PIXEL ARRAYS STORED IN MEMORY

B02 BLUE	401	759	746	2962
B03 GREEN	401	759	746	2962
B04 RED	401	759	746	2962
B08 NIR	401	759	746	2962

IN-MEMORY DATA

2 acquisition dates

4 bands per date

512 × 512 values / band

2,097,152 total values

Float arrays ready for analysis

- Data structure: bandData[date][band][pixel].
- Each band array stores 262,144 pixel values.
- The sample shown is row 0, column 0.
- Values are retained in memory for later NDWI/NDVI processing.

NDWI & NDVI Calculation

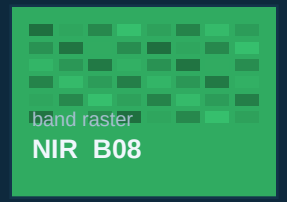
Both indices are calculated independently for the two image dates.

PROCESSING VISUAL

ILLUSTRATIVE · NOT ACTUAL OUTPUT

SPECTRAL BANDS · INDEX VALUES







NDWI

(Green · NIR) / (Green + NIR)

Water-sensitive index

NDVI

(NIR · Red) / (NIR + Red)

Vegetation-sensitive index

NDWI

(Green – NIR) / (Green + NIR)

NDVI

(NIR – Red) / (NIR + Red)

Reported valid pixels:

NDWI: 262,144 per date

NDVI: 262,144 per date

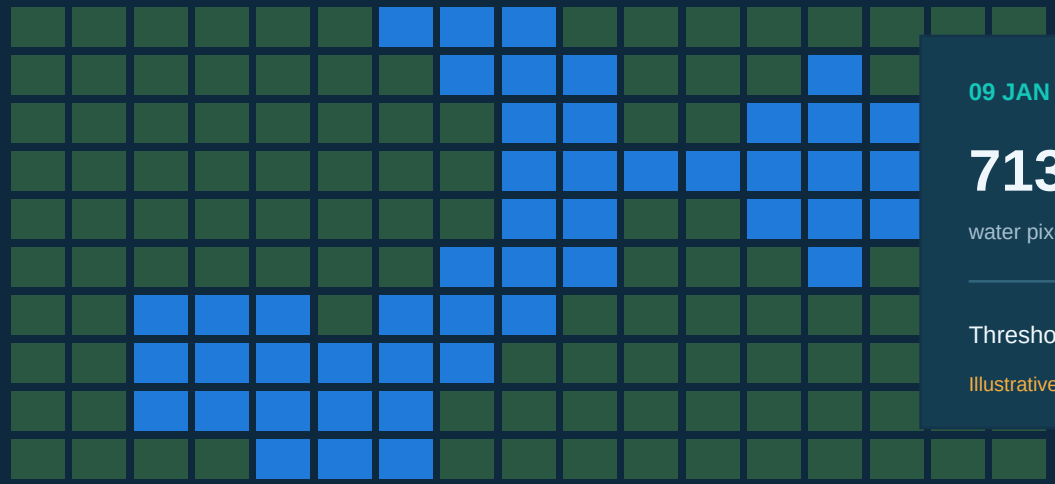
Pre-flood Water Map — 09 January 2019

A binary water map was generated from NDWI using a threshold of 0.0.

PROCESSING VISUAL

ILLUSTRATIVE · NOT ACTUAL OUTPUT

BINARY WATER CLASSIFICATION



Reported output

713

Water pixels

261,431

Non-water pixels

Threshold: NDWI > 0

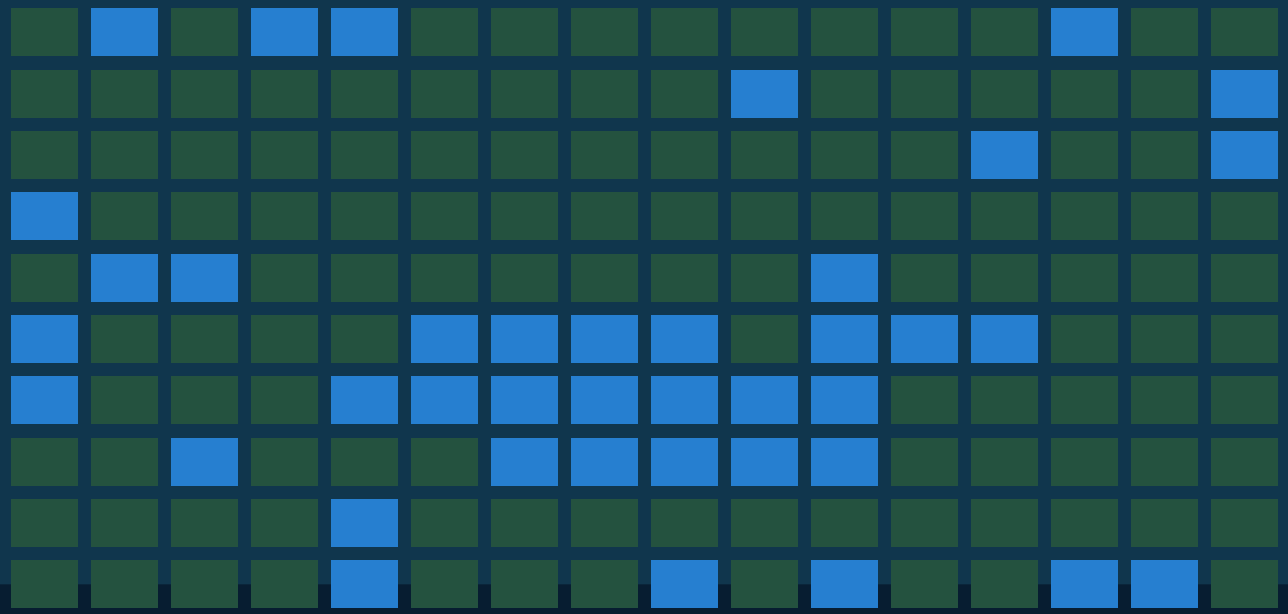
Map values: 1 water | 0 non-water

255 reserved for NoData

Post-flood Water Map — 14 January 2019

The same NDWI threshold workflow was applied to the second acquisition date.

BINARY WATER CLASSIFICATION



REPORTED OUTPUT

682

Water pixels

261,462 non-water pixels

Threshold: NDWI > 0

Map: 1 = water | 0 = non-water

255 = NoData

What the Current Results Tell Us

The generated maps are date-specific water classifications; flood-event status still needs independent verification.

CURRENT MIDTERM OUTPUTS

01 **Image reading completed**

All 8 single-band Sentinel-2 images were opened and their pixel values loaded into memory.

02 **Input validation completed**

Image dimensions (512 × 512), file opening, raster grid data and CRS were checked.

03 **Image preprocessing completed**

Pixel values were checked and invalid NDWI denominator cases (Green + NIR = 0) were handled.

04 **Band storage completed**

All four bands for both acquisition dates were stored in arrays for further analysis.

05 **NDWI and NDVI calculated**

Both spectral indices were calculated for the two dates using the selected Sentinel-2 bands.

06 **Pre-flood and post-flood water maps generated**

Separate binary water maps were created using the NDWI threshold (NDWI > 0).

Flood differencing, connected-region analysis, area calculation and severity classification are planned for the final phase.

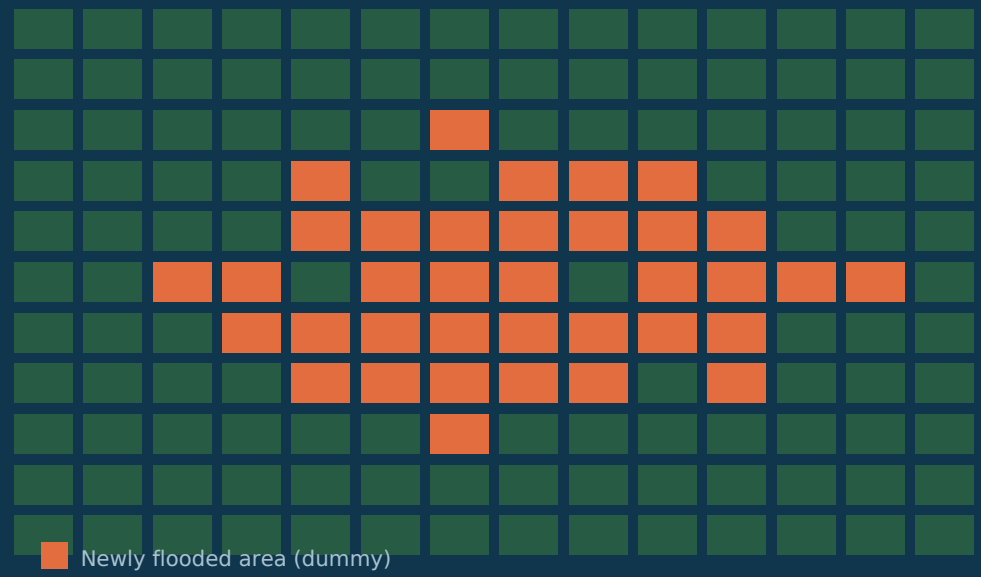
Midterm Work Completed

A quick overview of the processing completed up to the midterm stage.

COMPLETED UP TO MIDTERM

- Collected Sentinel-2 images for 09 and 14 January 2019.
- Read all 8 single-band images.
- Validated image dimensions, file opening, grid data and CRS.
- Checked pixel validity and NDWI denominator values.
- Stored band values and calculated NDWI / NDVI.
- Generated pre-flood and post-flood water maps.

ILLUSTRATIVE FINAL FLOOD MAP

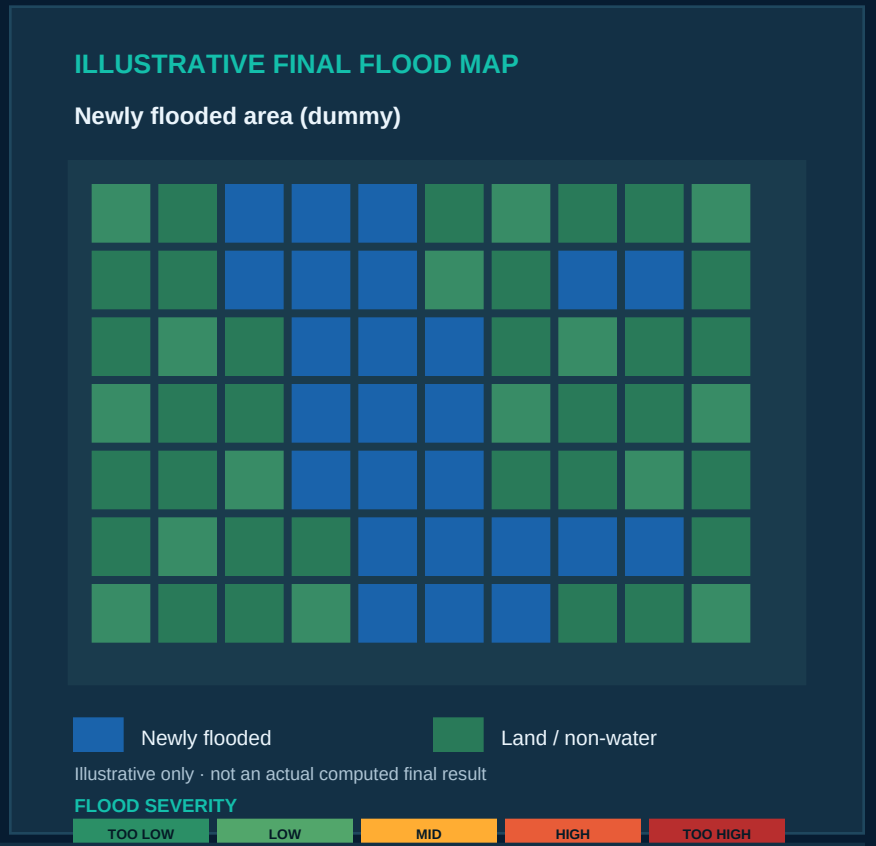


Illustrative only — not an actual computed final result

What I Will Do in the Final

A step-by-step roadmap for producing the final flood-extent map.

1. **Compare pre- and post-flood water maps**
   Use the aligned water maps of 09 January and 14 January 2019.
2. **Identify newly flooded pixels**
   Pre-flood = 0 AND post-flood water = 1 · newly flooded pixel = 1.
3. **Apply morphological operations**
   Remove isolated noise and refine the newly flooded regions.
4. **Analyze connected regions using BFS / DFS**
   Identify connected flooded areas and separate flooded regions.
5. **Find the region with the largest flooded area**
   Use BFS / DFS to compare regions and report which region contains the most flooded pixels.
6. **Calculate flooded pixels and area**
   Count flooded pixels in each region and estimate the total flooded area.
7. **Classify flood severity**
   Use flooded-pixel / area results: Too Low | Low | Mid | High | Too High.
8. **Generate the final flood map**
   Display newly flooded regions and summarize the detected flood extent.



FINAL FLOOD PIXEL = 1 · PRE-FLOOD = 0 AND POST-FLOOD WATER = 1

THANK YOU | Questions & Discussion